Schedule

- Case exam 2: May 29th Thursday at 17:30-20:30
  - Hypothesis testing
  - Multiple regression
- No class on Thursday
- Final exam: June 13th

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Agenda

- Short review
- Logistic regression
- Model selection

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Summary of Multiple LR

The sample: $\{(x_{1i}, x_{2i}, \ldots, x_{ki}, y_i)\}_{i=1}^{n}$

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k}_{\text{True regression line}} + \varepsilon \qquad \varepsilon \sim N(0, \sigma^2)$$

- Significance of LR model
  - F-test for the whole model
  - t-tests for individual independent variables.

$H_0: \beta_i = 0$ for all i
$H_a: \beta_i \neq 0$ for at least one i

$H_0: \beta_i = 0$
$H_a: \beta_i \neq 0$

- Percentage of variability explained by the LR model
  - R^2 : no reference to df
  - Adjusted R^2 : adjusted according to df so as to minimize MSE
- Other criteria: C_p

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Verifying the assumptions: Diagnostic Plots

$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \ldots + \beta_k X_{ki} + \varepsilon_i$
$\varepsilon_i \sim N(0, \sigma^2)$

		Assumptions on ε_i:
1) e_i^* (or e_i) versus x_i	⟷	1) ε_i's (so $E(\varepsilon_i)$ and $V(\varepsilon_i)$) do not depend on x_i.
2) e_i^* (or e_i) versus y_i	⟷	2) ε_i's (so $E(\varepsilon_i)$ and $V(\varepsilon_i)$) do not depend on y_i.
3) normal probability plot of e_i^* (or e_i)	⟷	3) ε_i's are normal.
4) e_i^* (or e_i) versus i	⟷	4) ε_i's are independent over time. i.e., do not depend on i.

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Verifying the assumptions: Diagnostic Plots

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \ldots + \beta_k X_{ki} + \varepsilon_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

5) $\hat{y}_i$ versus y_i ⟷ 5) close to a 45° line??

6) y_i versus x_i ⟷ 6) an observation with large influence

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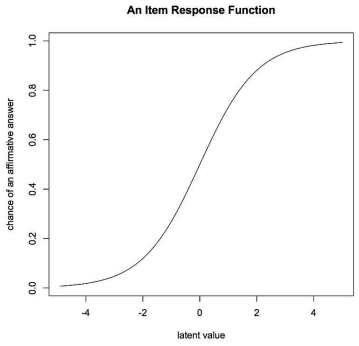
Logistic regression

$H_0: \beta_1 = 0$
$H_a: \beta_1 \neq 0$

- Y: binary variable
- Logit response function: $E(Y) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$

$$E(Y) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- Odds ratio: $\frac{E(Y)}{1 - E(Y)} = e^{(\beta_0 + \beta_1 x)}$



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Example: Challenger O-Ring Data

- The Space Shuttle Challenger exploded 73 seconds after lift-off on January 28th, 1986. The disaster claimed the lives of all seven astronauts on board, including school teacher Christa McAuliffe. The details surrounding this disaster were very involved.
- For the purposes of this analysis, it is sufficient to point out that engineers who manufactured the large boosters that launched the rocket were aware of the possible failures that could happen during cold temperatures. They tried to prevent the launch, but were ultimately ignored, and disaster ensued.
- The main concern of engineers in launching the Challenger was the evidence that the large O-rings sealing the several sections of the boosters could fail in cold temperatures.

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Data

date	temp	fail
4.12.1981	66	0
11.12.1981	70	1
3/22/82	69	0
11.11.1982	68	0
4.04.1983	67	0
6/18/83	72	0
8/30/83	73	0
11/28/83	70	0
2.03.1984	57	1
4.06.1984	63	1
8/30/84	70	1
10.05.1984	78	0
11.08.1984	67	0
1/24/85	53	2
4.12.1985	67	0
4/29/85	75	0
6/17/85	70	0
7/29/85	81	0
8/27/85	76	0
10.03.1985	79	0
10/30/85	75	2
11/26/85	76	0
1.12.1986	58	1

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Logistic Model

The probability of at least one o-ring failing during a shuttle launch based on the known outside temperature at the time of launch is given by the following logistic regression model.

$$P(Y_i = 1|x_i) = \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}} = \pi_i$$

In this model, for each previous shuttle launch i:

- $Y_i = 1$ denotes at least one o-ring failing for the given launch,
- $Y_i = 0$ denotes no o-rings failing (successful launch), and
- x_i denotes the outside temperature in degrees Fahrenheit at the time of the launch.

If β_1 is zero in the above model, then x_i (temperature) provides no insight about the probability of a failed O-ring. If not zero however, then temperature plays an important role in the probability of o-rings failing. Using a significance level of $\alpha = 0.05$ we will test the below hypotheses about β_1.

$$H_0 : \beta_1 = 0$$
$$H_a : \beta_1 \neq 0$$

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Fitting the Model

The estimates of the coefficients β_0 and β_1 for the above logistic regression model and data are shown below.

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	15.04	7.379	2.039	0.04148
temp	-0.2322	0.1082	-2.145	0.03196

(Dispersion parameter for binomial family taken to be 1)

Null deviance:	28.27 on 22 degrees of freedom
Residual deviance:	20.32 on 21 degrees of freedom

This gives the estimated model for π_i as

$$P(Y_i = 1|x_i) \approx \frac{e^{15.043-0.232x_i}}{1 + e^{15.043-0.232x_i}} = \hat{\pi}_i$$

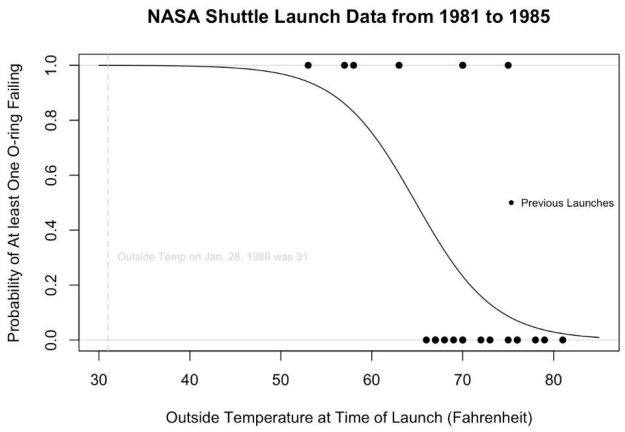
where $b_0 = 15.043$ is the value of the `(Intercept)` which estimates β_0 and $b_1 = -0.232$ is the value of `temp` which estimates β_1.

Importantly, the p-value for the test of `temp` shows a significant result $(p = 0.0320)$ giving sufficient evidence to conclude that $\beta_1 \neq 0$. The outside temperature at the time of launch effects the probability of at least one o-ring failure during the launch.

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date	temp	fail
4.12.1981	66	0
11.12.1981	70	1
3/22/82	69	0
11.11.1982	68	0
4.04.1983	67	0
6/18/83	72	0
8/30/83	73	0
11/28/83	70	0
2.03.1984	57	1
4.06.1984	63	1
8/30/84	70	1
10.05.1984	78	0
11.08.1984	67	0
1/24/85	53	2
4.12.1985	67	0
4/29/85	75	0
6/17/85	70	0
7/29/85	81	0
8/27/85	76	0
10.03.1985	79	0
10/30/85	75	2
11/26/85	76	0
1.12.1986	58	1

Visualizing the Model



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Conclusion

Since the temperature being zero is not really realistic for this model, the value of e^{b_0} is not interpretable. However, the value of $e^{b_1} = e^{-0.232} \approx 0.79$ shows that the odds of the o-rings failing for a given launch decreases by a factor of 0.79 for every 1° F increase in temperature. Said differently, the odds of an o-ring failure during launch decrease by 21% (1-0.79) for every 1° F increase in temperature. (Also, from the reverse perspective, every 1° F decrease in temperature increases the odds of a failed o-ring by a factor of $e^{0.232} \approx 1.26$.) The Challenger shuttle was launched at a temperature of 31° F. By waiting until 53° F, the odds of failure would have been decreased by a factor of $e^{-0.232(53-31)} \approx 0.006$, a 99.4% reduction in the odds of an o-ring failure!

To state it more clearly, for a temperature of 31° F our model puts the probability of a failure at

$$P(Y_i = 1|x_i) \approx \frac{e^{15.043-0.232 \cdot 31}}{1 + e^{15.043-0.232 \cdot 31}} = \hat{\pi}_i$$

Code

which, using R to do this calculation we get $\hat{\pi}_i \approx 0.99961$. This shows that an O-ring failure was sure to happen when the launch temperature was that cold (31°F).

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Model Selection

- There are many situations in which a large number of independent variables have been measured, and we need to decide which of them to include in a model.
- This is the problem of **model selection**, and it is a difficult one.
- In practice, model selection often proceeds by ad hoc methods, guided by whatever physical intuition may be available.
- We will not attempt a complete discussion of this extensive and difficult topic. Instead, we will be content to state some basic principles and to present some examples.

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Model Selection and Occam's Razor

- Good model selection rests on this basic principle known as Occam's razor:
  "The best scientific model is the simplest model that explains the observed data."
- In terms of linear models, Occam's razor implies the **Principle of Parsimony**:
  "A model should contain the smallest number of variables necessary to fit the data."

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Some Exceptions

1. A linear model should always contain an intercept, unless physical theory dictates otherwise.
2. If a power X^n of a variable is included in the model, all lower powers $X, X^2, \ldots, X^{n-1}$ should be included as well, unless physical theory dictates otherwise.
3. If a product $x_i x_j$ of two variables is included in a model, then the variables x_i and x_j should be included separately as well, unless physical theory dictates otherwise.

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Notes

- Models that include only the variables needed to fit the data are called **parsimonious** models. Much of the practical work of multiple regression involves the development of parsimonious models.
- Adding a new variable to a model can substantially change the coefficients of the variables already in the model.

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Example

TABLE 8.4 Data overhead, speed, pause time, and link change rate for a mobile computer network

Speed (m/s)	Pause Time (s)	LCR (100/s)	Data Overhead (kB)
5	10	9.426	428.90
5	20	8.318	443.68
5	30	7.366	452.38
5	40	6.744	461.24
5	50	6.059	475.07
10	10	16.456	446.06
10	20	13.281	465.89
10	30	11.155	477.07
10	40	9.506	488.73
10	50	8.310	498.77
20	10	26.314	452.24
20	20	19.013	475.97
20	30	14.725	499.67
20	40	12.117	501.48
20	50	10.284	519.20
30	10	33.009	445.45
30	20	22.125	489.02
30	30	16.695	506.23
30	40	13.257	516.27
30	50	11.107	508.18
40	10	37.823	444.41
40	20	24.140	490.58
40	30	17.700	511.35
40	40	14.064	523.12
40	50	11.691	523.36

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Model 0

```
The regression equation is
Overhead = 368 + 3.48 LCR + 3.04 Speed + 2.29 Pause - 0.0122 Speed*Pause
           -0.1041 LCR^2 - 0.0313 Speed^2 - 0.0132 Pause^2

Predictor            Coef         SE Coef          T          P
Constant           367.96           19.40      18.96      0.000
LCR                 3.477           2.129       1.63      0.121
Speed               3.044           1.591       1.91      0.073
Pause              2.2924          0.6984       3.28      0.004
Speed*Pa          -0.01222         0.01534      -0.80      0.437
LCR^2             -0.10412         0.03192      -3.26      0.005
Speed^2           -0.03131         0.01906      -1.64      0.119
Pause^2           -0.01318         0.01045      -1.26      0.224

S = 5.72344           R-Sq = 97.2%          R-Sq(adj) = 96.1%
Analysis of Variance

Source           DF           SS           MS          F          P
Regression        7        19567.5       2795.4      85.33      0.000
Residual Error   17          556.9         32.8
Total            24        20124.3
```

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Model 1

The regression equation is
Overhead = 359 + 6.69 LCR + 0.777 Speed + 1.67 Pause − 0.156 LCR^2

Predictor	Coef	SE Coef	T	P
Constant	359.22	13.01	27.61	0.000
LCR	6.695	1.156	5.79	0.000
Speed	0.7766	0.2054	3.78	0.001
Pause	1.6729	0.1826	9.16	0.000
LCR^2	−1.5572	0.02144	−7.26	0.000

S = 6.44304 R-Sq = 95.9% R-Sq(adj) = 95.0%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	4	19294.1	4823.5	116.19	0.000
Residual Error	20	830.3	41.5		
Total	24	20124.3			

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Can a Variable Be Dropped?

- It often happens that one has formed a model that contains a large number of independent variables, and one wishes to determine whether a given subset of them may be dropped from the model without significantly reducing the accuracy of the model.
- Assume that we know that the model

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \beta_{k+1} x_{k+1i} + \cdots + \beta_p x_{pi} + \varepsilon_i$$

is correct. We will call this the **"full" model**.

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Null Hypothesis

- We wish to test the null hypothesis

$$H_0: \beta_{k+1} = \cdots = \beta_p = 0$$

- If H_0 is true, the model will remain correct if we drop the variables $X_{k+1}, \ldots, X_p$, so we can replace the full model with the following **reduced model**:

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \varepsilon_i$$

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$H_0: \beta_{k+1} = \cdots = \beta_p = 0$

Test Statistic

- To develop a test statistic for H_0, we begin by computing the error sums of squares for both the full and reduced models.
- We call them SS_{full} and $SS_{reduced}$, respectively.
- The number of degrees of freedom for SS_{full} is $n - p - 1$, and for $SS_{reduced}$ is $n - k - 1$.

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$H_0: \beta_{k+1} = \cdots = \beta_p = 0$

Test Statistic

- The test statistic is

$$f = \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}})/(p-k)}{\text{SSE}_{\text{full}}/(n-p-1)}$$

- If H_0 is true, both numerator and denominator are estimates of σ^2, so f tends to be close to 1.

  If H_0 is false, the quantity $\text{SSE}_{\text{reduced}}$ tends to be larger, so f tends to be larger.

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Example: F statistic

Now we can compute the F statistic. Using Equation (8.17), we compute

$$f = \frac{(830.3 - 556.9)/(7-4)}{556.9/17} = 2.78$$

The null distribution is $F_{3,17}$. From the F table (Table A.8 in Appendix A), we find that $0.05 < P < 0.10$. According to the 5% rule of thumb, since $P > 0.05$, the reduced model is plausible, but only barely so. Rather than settle for a barely plausible model, it is wise to explore further, to look for a slightly less reduced model that has a larger P-value.

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Model 2

The regression equation is
Overhead = 373 + 4.80 LCR + 1.99 Speed + 1.45 Pause − 0.123 LCR^2
−0.0212 Speed^2

Predictor	Coef	SE Coef	T	P
Constant	372.60	16.93	22.00	0.000
LCR	4.799	1.935	2.48	0.023
Speed	1.993	1.023	1.95	0.066
Pause	1.4479	0.2587	5.60	0.000
LCR^2	−0.12345	0.03400	−3.63	0.002
Speed^2	−0.02120	0.01746	−1.21	0.240

S = 6.36809 R−Sq = 96.2% R−Sq(adj) = 95.2%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	5	19353.8	3870.8	95.45	0.000
Residual Error	19	770.5	40.6		
Total	24	20124.3			

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Model 3

The regression equation is
Overhead = 345 + 6.484 LCR + 0.707 Speed + 2.85 Pause − 0.145 LCR^2
− 0.0183 Pause^2

Predictor	Coef	SE Coef	T	P
Constant	345.42	13.19	26.20	0.000
LCR	6.484	1.050	6.17	0.000
Speed	0.7072	0.1883	3.76	0.001
Pause	2.8537	0.5337	5.35	0.000
LCR^2	−0.14482	0.01996	−7.25	0.000
Pause^2	−0.018334	0.007879	−2.33	0.031

S = 5.83154 R−Sq = 96.8% R−Sq(adj) = 95.9%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	5	19478.2	3895.6	114.55	0.000
Residual Error	19	646.1	34.0		
Total	24	20124.3			

This model looks good, at least at first. All the variables have small P-values. We'll compute the F statistic to see if this model is plausible. The value of $SSE_{reduced}$ in this model is 646.1. The number of independent variables is $k = 5$. The value of the F statistic, using Equation (8.17), is therefore

$$f = \frac{(646.1 - 556.9)/(7 - 5)}{556.9/17} = 1.36$$

P value>0.10

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Comments

- The method we have just described is very useful in practice for developing parsimonious models by removing unnecessary variables.
- However, the conditions under which it is formally valid are seldom met in practice.

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Why Conditions May Not Be Met

1. It is rarely the case that the full model is correct; there will be nonrandom quantities that affect the value of the dependent variable *y* that are not accounted for by the independent variables.
2. For the method to be formally valid, the subset of variables to be dropped must be determined independently of the data. This is usually not the case.

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Comments

- More often, a large model is fit, some of the variables are seen to have fairly large *P*-values, and the *F* test is used to decide whether to drop them from the model.
- This is a useful technique in practice, but, like most methods of model selection, it should be seen as an informal tool rather than a rigorous theory-based procedure.
- As such, it is possible to find a different final model that might have been just as good as the one we actually found.
- In practice, there are often many models that fit the data about equally well; there is no single "correct" model.

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Methods to find the "best" regression model

- All possible regressions
- Stepwise regression (partial F-tests)
- Forward selection
- Backward elimination

- Performed by statistical packages.

- There is not one specific rule to choose the "best" model. You should use your judgement, given all the criteria...

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Best Subsets Regression

- Assume that there are *p* independent variables, $x_1, x_2, \ldots, x_p$, that are available to be put in the model.
- Let's assume that we wish to find a good model that contains exactly four independent variables.
- We can simply fit every possible model containing four of the variables, and rank them in order of their goodness-of-fit, as measured by the coefficient of determination, R^2.
- The subset of four variables that yield the largest value R^2 of is the "best" subset of size four.

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Best Subsets Regression

- The process can be repeated for subsets of other sizes, finding the best subsets of size $1,2, \ldots, p$.
- These best subsets can be examined to see which provides a good fit, while being parsimonious.
- The best subsets procedure is computationally intensive.
- When there are a lot of potential independent variables, there are a lot of models to fit. However, for most data sets, we can rely on technology to do our calculations.

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Determining "Best"

- Methods have been developed to choose a single model, presumably the "best" of the "best."
- We describe two of them here, with a caution not to take them too seriously.
- Note that if we simply choose the model with the highest value of R^2, we will always pick the one that contains all the variables, since the value of R^2 necessarily increases as the number of variables in the model increases.
- The methods for selecting a model involve statistics that adjust the value of R^2, so as to eliminate this feature

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Determining "Best" – Method 1

- The first is the adjusted R^2. Let *n* denote the number of observations, and let *k* denote the number of independent variables in the model. The adjusted R^2 is defined as follows:

$$\text{Adjusted } R^2 = R^2 - \left(\frac{k}{n-k-1}\right)(1-R^2)$$

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Determining "Best" – Method 2

- The adjusted R^2 is always smaller than R^2, since a positive quantity is subtracted from R^2.
- As the number of variables *k* increases, R^2 will increase, but the amount subtracted from it will increase as well.
- The value of *k* for which the value of adjusted R^2 is a maximum can be used to determine the number of variables in the model, and the best subset of that size can be chosen as the model.

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Determining "Best" – Method 3

- Another commonly used statistic is **Mallows' C_p**.
- To compute this quantity, let *n* be the number of observations, let *p* be the total number of independent variables under consideration, and let *k* be the number of independent variables in a subset.
- As before, let SSE_{full} denote the error sum of squares for the full model containing all *p* variables, and let $SSE_{reduced}$ denote the error sum of squares for the model containing only the subset of *k* variables.

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Determining "Best" – Method 4

- Mallows' C_p is defined as:

$$C_p = \frac{(n-p-1)\text{SSE}_{\text{reduced}}}{\text{SSE}_{\text{full}}} - (n-2k-2)$$

- For models that contain as many independent variables as necessary, the value of C_p is supposed to be approximately equal to the number of variables, including the intercept, in the model.
- To choose a single model, one can either choose the model with the minimum value of C_p, or one can choose the model in which the value of C_p is closest to the number of independent variables in the model.

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Stepwise Regression

- This is the most widely use model selection technique.
- Its main advantage over best subsets regression is that it is less computationally intensive, so it can be used in situations where there are a very large number of candidate independent variables and too many possible subsets for every one of them to be examined.
- The version of stepwise regression that we will describe is based on the *P*-values of the *t* statistics for the independent variables. An equivalent version is based on the *F* statistic (which is the square of the *t* statistic).

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Stepwise Regression 2

- Before running the algorithm, the user chooses two threshold *P*-values, α_{in} and α_{out}, with $\alpha_{\text{in}} \leq \alpha_{\text{out}}$.
- The stepwise regression procedure begins with a step called a **forward selection** step, in which the independent variables with smallest *P*-value is selected, provided that $P < \alpha_{\text{in}}$.
- This variable is entered in the model, creating a model with a single independent variable.

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More on Stepwise Regression 1

- In the next step, the remaining variables are examined one at a time as candidates for the second variable in the model. The one with the smallest *P*-value is added to the model, again provided that $P < \alpha_{\text{in}}$.
- Now, it is possible that adding the second variables to the model increased the *P*-value of the first variable. In the next step, called a **backward elimination** step, the first variable is dropped from the model if its *P*-value has grown to exceed the value α_{out}.

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More on Stepwise Regression 2

- The algorithm continues by alternating forward selection steps with backward eliminations steps.
- The algorithm terminates when no variables meet the criteria for being added to or dropped from the model.

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Notes on Model Selection

- When there is little or no physical theory to rely on, many different models will fit the data about equally well.
- The methods for choosing a model involve statistics, whose values depend on the data. Therefore, if the experiment is repeated, these statistics will come out differently, and different models may appear to be “best.”
- Some or all of the independent variables in a selected model may not really be related to the dependent variable. Whenever possible, experiments should be repeated to test these apparent relationships.
- Model selection is an art, not a science.

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Summary

In this chapter, we learned about

- multiple regression models.
- estimating the coefficients.
- checking assumptions in multiple regression.
- confounding and collinearity.
- model selection.